⌈Def. Given two tangent vectors V_p and W_p at $P\in\mathbb{R}^3$, define

$$V_p\times W_p \overset{\text{def}}{=} \begin{bmatrix}1&0&0\\V_1&V_2&V_3\\W_1&W_2&W_3\end{bmatrix}\cdot(e_1)_p+\begin{bmatrix}0&1&0\\V_1&V_2&V_3\\W_1&W_2&W_3\end{bmatrix}\cdot(e_2)_p+\begin{bmatrix}0&0&1\\V_1&V_2&V_3\\W_1&W_2&W_3\end{bmatrix}\cdot(e_3)_p.$$

It is called the Cross Product of V_p and W_p.

⌈Prop. $V_p\times W_p=0$ if and only if V_p and W_p are linearly dependent.

Proof. Suppose that $V_p\times W_p=0$. Then, for any $(u_1,u_2,u_3)\in\mathbb{R}^3$,

$$\det\begin{bmatrix}u_1&u_2&u_3\\V_1&V_2&V_3\\W_1&W_2&W_3\end{bmatrix}=0.$$

So, taking $(u_1,u_2,u_3)\in\mathbb{R}^3\setminus\operatorname{Span}\{V,W\}$, we obtain the result.
(If V and W are linearly independent, then the determinant cannot be 0.)

Conversely, if V_p and W_p are linearly dependent, then regardless of the first row, $\det=0$ $\square$

⌈Lemma. Given V_p and W_p in $T_p(\mathbb{R}^3)$, $V_p\times W_p$ is orthogonal to both V_p and W_p, and

$$\|V_p\times W_p\|^2=(V_p\cdot V_p)\cdot(W_p\cdot W_p)-(V_p\cdot W_p)^2.$$

In other say, $\|V\times W\|=\|V\|\cdot\|W\|\cdot\sin\theta$

Proof. By the linearity of the determinant,

$$V\cdot(V\times W)=\begin{vmatrix}V_1&V_2&V_3\\V_1&V_2&V_3\\W_1&W_2&W_3\end{vmatrix}=0 \quad\text{and}\quad W\cdot(V\times W)=\begin{vmatrix}W_1&W_2&W_3\\V_1&V_2&V_3\\W_1&W_2&W_3\end{vmatrix}=0.$$

Therefore, the orthogonality is proved. And, to show the second assertion,

$$(V\cdot V)\cdot(W\cdot W)-(V\cdot W)^2=(V_1^2+V_2^2+V_3^2)(W_1^2+W_2^2+W_3^2)-(V_1W_1+V_2W_2+V_3W_3)^2$$

$$=\sum_{1\le i,j\le 3}V_i^2\cdot W_j^2-\Big(\sum_{1\le i\le j\le 3}V_iW_iV_jW_j\Big)$$

$$=\sum_{1\le i,j\le 3}V_i^2\cdot W_j^2-\Big(\sum_{i=1}^{3}V_i^2W_i^2+2\cdot\sum_{1\le i<j\le 3}V_iW_iV_jW_j\Big)$$

$$=\sum_{1\le i<j\le 3}V_i^2\cdot W_j^2-2\cdot\sum_{1\le i<j\le 3}V_iW_iV_jW_j$$

Meanwhile,

$$\|V\times W\|^2=(V_2W_3-V_3W_2)^2+(V_1W_3-V_3W_1)^2+(V_1W_2-V_2W_1)^2$$

$$\begin{aligned}&=V_2^2W_3^2+V_3^2W_2^2-2V_2W_2V_3W_3\\&\quad+V_1^2W_3^2+V_3^2W_1^2-2V_1W_1V_3W_3\\&\quad+V_1^2W_2^2+V_2^2W_1^2-2V_1W_1V_2W_2\end{aligned}$$

These are same. $\square$